

TRIBE 1 · THE BRAIN

The ML / AI Engineer

Computer Science / AI Foundation

Core Strength:

Semantic Competence & Representation

- Vision-Language Models (VLMs)
- Diffusion / ACT Trajectory Chunks
- Latent World Models (JEPAs)

The Critical Blindspot:

The Digital Sandbox Illusion—assuming simulation proof guarantees safety and software exceptions are harmless.

Semantic Proposals (p_t)



TRIBE 2 · THE NERVOUS SYSTEM

The Embedded / ECE Engineer

Silicon & Real-Time Systems

Core Strength:

Silicon Privilege & Multi-Rate IPC

- Microsecond Clocks & P_{99} Metrology
- Zero-Copy DMA & Shared SRAM
- Hardware Peripheral Bus Firewalls

The Critical Blindspot:

The Static Automation Illusion—treating systems as rigid, static loops unable to parse unstructured open worlds.

Multi-Rate IPC & Watchdogs



TRIBE 3 · THE BODY & CONTROL

The Robotics Engineer

Dynamics, Invariants & Control

Core Strength:

Physical Laws & Safety Envelopes

- Work-Energy & Inertia $\mathbf{M}(\mathbf{q})$
- Control Barrier Functions ($h(x) \geq 0$)
- Thermal Limits (I^2t) & Jerk Limits

The Critical Blindspot:

The Closed-World Illusion—distrusting learned models as black boxes; failing when lighting or physical objects deviate.

1 kHz CBF Safe Sets ($h(x) \geq 0$)



THE PHYSICAL AI SYSTEMS SYNTHESIS

Bridging the Brain, Nervous System, and Body across the Proposal-Permission Boundary

1. Unverified Proposals (Brain)

High-capacity VLMs & Diffusion ACT emit candidate action chunks (p_t) on Linux MPU.

2. Real-Time Transport (Nerves)

Lock-free SRAM ring buffers and hardware watchdogs bound tail latency (P_{99}).

3. Physical Invariants (Body)

1 kHz MCU projects proposals onto safe sets $\Pi_{\mathcal{U}_{\text{safe}}}(p_t)$ via CBFs before gate drive.

Universal Definition of Success: Open-World Semantic Competence AND Strict Physical Invariant Survival